MINI PROJECT REPORT

ON

ROADSENSE AI

Submitted By

ARJUN BAIJU (CEC25MCA-2011)

under the esteemed guidance of

Mr.Gokul Viswanath

In partial fulfillment of the requirements for the award of the degree
in
Master of Computer Applications
of
APJ Abdul Kalam Technological University



OCTOBER 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING, Pallippuram P O, Cherthala,
Alappuzha Pin: 688541,

http://www.cectl.ac.in

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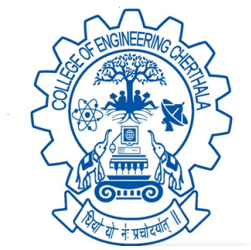
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C E R T I F I C A T E

This is to certify that, the Mini project report titled ***ROADSENSE AI*** is a bonafide record of the **20MCA246 Mini Project** presented by **ARJUN BAIJU(CEC25MCA-2011)** Third Semester Master of Computer Application student, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree, **Master of Computer Application** of **APJ Abdul Kalam Technological University during the academic year 2026-2027**.

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I would like to thank my dear friends for their cooperation and encouragement throughout the project work, without which I could not have completed the project successfully. Thank you all for your love and understanding.

DECLARATION

I hereby declare that the project **"ROADSENSE AI "** is a bonafide work carried out by me during the academic year 2026–2027 under the guidance of **Mr.Gokul Viswanath** , Assistant Professor, at College of Engineering, Cherthala. This report has not been submitted previously for the award of any degree, diploma, fellowship, or any other similar title or recognition at any other university.

ARJUN BAIJU

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22/10/2026

ABSTRACT

The rapid advancement of Artificial Intelligence and Computer Vision technologies has created a growing demand for intelligent transportation systems that can improve road safety and assist drivers in making timely decisions. Traditional methods of monitoring traffic signs rely entirely on human observation, which can be affected by distractions, fatigue, poor weather conditions, and limited visibility, leading to traffic violations and road accidents. To address these challenges, this project proposes RoadSense AI, a real-time Traffic Sign Detection and Recognition System that utilizes the YOLOv8 (You Only Look Once Version 8) object detection algorithm to accurately identify traffic signs from live video captured through a webcam. The system processes each video frame using a trained deep learning model and instantly detects traffic signs with high accuracy and low inference time. Once a traffic sign is recognized, the application displays the detected sign, confidence score, driver assistance message, inference time, frames per second (FPS), and detection count through a user-friendly web dashboard. To further enhance driver awareness, the system integrates a voice assistance module that announces the detected traffic sign in real time, enabling drivers to respond quickly and appropriately. The application is developed using Python, Flask, OpenCV, PyTorch, and the Ultralytics YOLOv8 framework, while the frontend is designed using HTML, CSS, Bootstrap, and JavaScript to provide an interactive and responsive interface. The current implementation is capable of detecting ten commonly used traffic signs, including Stop, No Entry, No Stopping, Parking, Speed Limit 40, Road Work, Roundabout, No U-Turn, No Overtaking, and Hump. By combining deep learning, computer vision, and web technologies, the proposed system provides a reliable, efficient, and intelligent solution for real-time traffic sign recognition. Furthermore, the project serves as a foundation for future Advanced Driver Assistance Systems (ADAS), where additional capabilities such as lane detection, traffic light recognition, pedestrian detection, vehicle detection, and autonomous driving assistance can be incorporated to further improve road safety and intelligent transportation.

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Chapter 1

INTRODUCTION

Road safety is one of the major concerns in modern transportation systems, where drivers must continuously observe and respond to various traffic signs while driving. However, traditional driving relies entirely on human observation, which can be affected by distractions, fatigue, poor weather conditions, and low visibility. Missing important traffic signs such as Stop, No Entry, or Speed Limit signs can lead to traffic violations, accidents, and loss of life. Artificial Intelligence and Computer Vision provide an efficient approach for developing intelligent traffic monitoring systems. Deep learning algorithms, particularly the YOLO (You Only Look Once) object detection model, enable fast and accurate recognition of traffic signs from live video streams. By integrating AI into driver assistance systems, traffic signs can be detected automatically, helping drivers make timely and informed decisions while driving. To further improve driver awareness, the proposed system incorporates a voice assistance module that announces the detected traffic sign in real time. The system captures live video using a webcam, processes each video frame through a trained YOLOv8 model, and displays the detected traffic sign, confidence score, driver assistance message, inference time, and other performance metrics on a web-based dashboard. The voice alert system ensures that drivers receive immediate audio notifications, even when they are unable to observe the display. The RoadSense AI system provides a reliable platform for real-time traffic sign detection and recognition by combining deep learning, computer vision, and web technologies. The application assists drivers by improving awareness of road signs, reducing the chances of missing critical traffic information, and enhancing overall road safety.

Chapter 2

PROBLEM STATEMENT

2.1 Problem Statement

RoadSense AI aims to develop an intelligent real-time traffic sign detection and recognition system using the YOLOv8 deep learning algorithm. The system will capture live video through a webcam or vehicle-mounted camera and process the video frames to automatically detect and identify traffic signs. The detected signs will be displayed with their corresponding bounding boxes and confidence scores, allowing the system to recognize important traffic signs such as stop, no entry, no stopping, no overtaking, speed limit, road work, and roundabout signs. The system is designed to provide fast and accurate detection with minimum delay, making it suitable for real-time driving assistance.

In addition to traffic sign detection, RoadSense AI will provide visual and voice-based alerts to assist the driver in understanding the detected signs and taking appropriate action. A Flask-based web dashboard will display the detected sign, confidence score, driver guidance, FPS, and other relevant system information. The system will also store detection details such as sign name, confidence value, driver instruction, and detection time in a MySQL database. Thus, RoadSense AI combines real-time detection, driver assistance, voice alerts, web-based monitoring, and detection history into a single traffic sign assistance system.

2.2 Objective

The primary objectives of the RoadSense AI project are:

1. To develop an AI-based traffic sign detection and recognition system.
2. To detect traffic signs in real time using the YOLOv8 object detection algorithm.
3. To provide immediate driver assistance messages based on detected traffic signs.
4. To generate voice alerts corresponding to each detected traffic sign.
5. To display real-time performance metrics including FPS, inference time, detection count, and GPU information.
6. To create a user-friendly web dashboard for monitoring live traffic sign detection.
7. To improve driver awareness and contribute towards safer road transportation.

Chapter 3

LITERATURE SURVEY

3.1 A Cascaded R-CNN with Multiscale Attention and Imbalanced Samples for Traffic Sign Detection

Traffic sign detection is challenging because signs can appear at different scales and may be affected by complex road environments. Zhang et al. (2020) proposed a cascaded R-CNN using Multiscale Attention, FPN, and CNN techniques. The method improves the detection of small traffic signs and handles imbalanced samples effectively. However, its complex architecture results in higher computational cost and slower inference.

3.2 Integrated Feature Pyramid Network with Feature Aggregation for Traffic Sign Detection

Detecting small traffic signs requires effective feature extraction at different scales. Tang et al. (2021) proposed an Integrated Feature Pyramid Network (IFA-FPN) with feature aggregation using Cascade R-CNN and FPN. The method improves feature extraction, mAP, and small sign detection. However, it has higher computational requirements and a complex model structure.

3.3 An Improved Faster R-CNN for Small Object Detection

Small objects are difficult to detect because they contain limited visual information. Cao et al. (2020) proposed an improved Faster R-CNN using CNN, RPN, and Feature Pyramid techniques. The method improves localization and detection of small objects. However, Faster R-CNN requires high computational resources and provides slower detection.

3.4 Traffic Sign Detection Using Feature Fusion and Contextual Information

Complex backgrounds and changing road conditions can affect traffic sign detection. Wang et al. (2021) proposed a method using Feature Fusion, Contextual Information, and CNN techniques. The approach improves detection accuracy and robustness by combining useful features. However, it requires higher computation and depends on scene context.

3.5 Traffic Sign Recognition Algorithm Based on Feature Pyramid Attention

Accurate traffic sign recognition requires effective feature extraction while reducing background interference. Su et al. (2021) proposed a Feature Pyramid Attention-based recognition algorithm using CNN and deep learning techniques. The method improves recognition accuracy and reduces background noise. However, it mainly focuses on recognition and requires higher computational resources.

3.6 Traffic Sign Detection and Recognition Using Multi-Scale Fusion and Prime Sample Attention

Traffic signs appear at different scales, making multi-scale feature extraction important for accurate detection. Cao et al. (2021) proposed a method using Multi-scale Fusion and Prime Sample

Attention with CNN. The approach improves detection at different scales and focuses on important samples. However, the model is complex and requires increased computational resources.

3.7 Improved Faster R-CNN Traffic Sign Detection Based on a Second Region of Interest and Highly Possible Regions Proposal Network

Accurate region proposals are important for detecting traffic signs, especially small signs. Shao et al. (2020) proposed an improved Faster R-CNN using a Second RoI and highly possible regions proposal network. The method improves region proposal and traffic sign detection accuracy. However, the two-stage architecture is computationally expensive and has slower inference.

3.8 Improved Traffic Sign Detection Algorithm Based on Faster R-CNN

Traffic sign detection can be affected by complex backgrounds and environmental conditions. Gao et al. (2022) proposed an improved Faster R-CNN using CNN, Feature Extraction, and RPN techniques. The method provides accurate detection and improved feature extraction. However, it has high computational cost and slower detection speed.

3.9 Improved Faster R-CNN Traffic Sign Detection Algorithm Based on Transformer

Detecting traffic signs in complex scenes requires both local and global feature information. Tian et al. (2023) proposed an improved Faster R-CNN using Transformer, CNN, and Attention Mechanism techniques. The method captures global features and improves detection in complex scenes. However, its architecture is complex and requires high computational resources.

3.10 Traffic Sign Detection Method Based on Improved SSD

Real-time traffic sign detection requires a balance between speed and accuracy. You et al. (2020) proposed an improved SSD-based method using CNN and Feature Extraction techniques. The approach provides fast detection and is suitable for real-time applications. However, its accuracy is lower for small objects and may decrease in complex scenes.

Table 3.1: Literature Survey - Comparison (1–5)

Paper Title (with Citation)	**Author's**	**Tools Used**	**Advantages**	**Disadvantages**
A cascaded R-CNN with multiscale attention and imbalanced samples for traffic sign detection [1]	Zhang et al. (2020)	Cascade R-CNN, Multiscale Attention, FPN, CNN	High accuracy for small traffic signs; handles imbalanced samples.	Complex architecture; slower inference.
Integrated feature pyramid network with feature aggregation for traffic sign detection [2]	Tang et al. (2021)	IFA-FPN, Feature Aggregation, Cascade R-CNN, FPN	Better feature extraction; improved mAP; detects small signs effectively.	Higher computational cost; complex model.
An improved faster R-CNN for small object detection [3]	Cao et al. (2020)	Faster R-CNN, CNN, RPN, Feature Pyramid	Improved localization and detection of small objects.	Slow detection speed; computationally expensive.
Traffic sign detection using feature fusion and contextual information [4]	Wang et al. (2021)	Feature Fusion, Contextual Information, CNN	Feature fusion improves accuracy and robustness.	Higher computation; depends on scene context.
Traffic sign recognition algorithm based on feature pyramid attention [5]	Su et al. (2021)	Feature Pyramid Attention, CNN, Deep Learning	Better recognition accuracy; reduces background noise.	Focuses mainly on recognition; higher computation.

Table 3.2: Literature Survey - Comparison (6–10)

Paper Title (with Citation)	**Author's**	**Tools Used**	**Advantages**	**Disadvantages**
Traffic sign detection and recognition using multi-scale fusion and prime sample attention [6]	Cao et al. (2021)	Multi-scale Fusion, Prime Sample Attention, CNN	Improves detection at different scales; focuses on important samples.	Complex model; increased computational requirements.
Improved faster R-CNN traffic sign detection based on a second region of interest and highly possible regions proposal network [7]	Shao et al. (2020)	Faster R-CNN, Second RoI, RPN, CNN	Improves region proposal; increases traffic sign detection accuracy.	Two-stage detection is computationally expensive; slower inference.
Improved traffic sign detection algorithm based on faster R-CNN [8]	Gao et al. (2022)	Faster R-CNN, CNN, Feature Extraction, RPN	Provides accurate detection; improves feature extraction.	High computational cost; slower detection speed.
Improved faster R-CNN traffic sign detection algorithm based on transformer [9]	Tian et al. (2023)	Transformer, Faster R-CNN, CNN, Attention Mechanism	Captures global features; improves detection in complex scenes.	High computational resources; complex architecture.
Traffic sign detection method based on improved SSD [10]	You et al. (2020)	Improved SSD, CNN, Feature Extraction	Fast detection; suitable for real-time applications.	Lower accuracy for small objects; performance decreases in complex scenes.

Chapter 4

PROPOSED SYSTEM

4.1 Solution

The RoadSense AI system is designed to provide a real-time, intelligent, and reliable traffic sign detection and recognition platform using Artificial Intelligence and Computer Vision technologies. The proposed system utilizes the YOLOv8 (You Only Look Once Version 8) object detection algorithm to identify traffic signs from live video captured through a webcam. The system processes each video frame in real time and accurately detects traffic signs while displaying the corresponding driver assistance message and voice alert. This approach improves driver awareness and helps reduce road accidents caused by missed or ignored traffic signs.

The application integrates a web-based dashboard developed using the Flask framework, allowing users to monitor live camera input and view detection results instantly. Whenever a traffic sign is detected, the system displays the sign name, confidence score, inference time, frames per second (FPS), GPU information, and detection count. In addition, a voice assistance module announces the detected traffic sign to ensure that drivers receive immediate audio feedback without constantly observing the display.

The key components of the system include a live webcam for video acquisition, a YOLOv8 deep learning model for traffic sign detection, OpenCV for image processing, Flask for backend communication, and HTML, CSS, Bootstrap, and JavaScript for developing the user interface. The proposed system combines these technologies to provide a fast, accurate, and user-friendly solu-

tion for real-time traffic sign recognition and driver assistance.

4.2 Feasibility Study

The feasibility study evaluates whether the proposed RoadSense AI system can be successfully designed, implemented, and deployed using the available resources. It analyzes different aspects of the project, including technical feasibility, social feasibility, schedule feasibility, operational feasibility, and economic feasibility.

The study ensures that the selected technologies are appropriate for implementing a real-time traffic sign detection system and that the proposed solution is practical for educational and intelligent transportation applications. The feasibility analysis confirms that the project can be completed successfully within the available time, budget, and hardware resources.

4.2.1 Technical Feasibility

The proposed system is technically feasible because all required software tools and hardware resources are readily available. The application is developed using Python, Flask, OpenCV, PyTorch, and the Ultralytics YOLOv8 framework, which provide efficient support for real-time object detection and web application development. The system utilizes a webcam for capturing live video and an NVIDIA GPU for accelerating deep learning inference. Since these technologies are open source and well documented, the project can be implemented efficiently using existing resources.

4.2.2 Social Feasibility

The proposed project is socially beneficial because it contributes to improving road safety by assisting drivers in recognizing traffic signs more effectively. The system provides both visual and voice-based alerts, enabling drivers to respond quickly to road signs even under poor visibility or distracted driving conditions. By reducing the possibility of missing important traffic signs, the project helps minimize traffic violations and accidents. Therefore, the proposed system is socially feasible and beneficial to society.

4.2.3 Schedule Feasibility

The development schedule for the RoadSense AI project is practical and achievable within the allocated project duration. The project activities, including requirement analysis, dataset preparation, model training, system design, implementation, testing, and deployment, are planned systematically. Proper task management and continuous testing ensure that the project can be completed within the specified academic schedule. Hence, the project is schedule feasible.

4.2.4 Operational Feasibility

The proposed system is designed with a simple and user-friendly interface that allows users to operate the application without requiring technical expertise. The web dashboard displays live camera footage, detected traffic signs, driver assistance messages, and system performance statistics in real time. The integrated voice assistant further improves usability by announcing detected traffic signs automatically. These features make the system reliable, efficient, and practical for real-time traffic monitoring applications. Therefore, the system is operationally feasible.

4.2.5 Economic Feasibility

The cost of developing the proposed system is relatively low because it utilizes open-source software technologies such as Python, Flask, OpenCV, PyTorch, and YOLOv8. The hardware requirements are also minimal, requiring only a standard computer with a webcam and an optional NVIDIA GPU for improved performance. The benefits of the system, including improved road safety, real-time driver assistance, and enhanced traffic sign recognition, outweigh the development costs. Therefore, the proposed system is economically feasible.

Chapter 5

SOFTWARE REQUIREMENT SPECIFICATION

5.1 Overall Description

The purpose of this document is to describe the functional and non-functional requirements of the RoadSense AI system. It defines the system objectives, functionalities, user interactions, and operational constraints. The proposed system provides a real-time traffic sign detection and recognition platform using Artificial Intelligence and Computer Vision technologies. By utilizing the YOLOv8 object detection algorithm, the application accurately detects traffic signs from live webcam video and provides visual as well as voice-based driver assistance to improve road safety and driver awareness.

5.1.1 Product Perspective

The RoadSense AI system is a web-based intelligent traffic sign detection application developed using deep learning and computer vision technologies. The application integrates a live webcam, YOLOv8 object detection model, backend server, and web dashboard to perform real-time traffic sign recognition. The system continuously captures video frames, processes them using the trained deep learning model, and displays the detection results along with driver assistance messages and performance statistics.

The system also integrates a voice assistance module that announces detected traffic signs in real time, allowing drivers to receive audio alerts while focusing on the road. The complete application

provides an efficient and user-friendly environment for intelligent traffic monitoring and driver assistance.

Core Technologies Used

- Frontend: HTML, CSS, Bootstrap, JavaScript
- Backend: Python,Flask
- Deep Learning Framework: Ultralytics YOLOv8
- Computer Vision: OpenCV
- Machine Learning Library: PyTorch
- Voice Assistant: pyttsx3
- Hardware: Webcam and NVIDIA GPU (Optional)

5.1.2 Software Interfaces

- **YOLOv8 Model:** Performs real-time traffic sign detection and recognition from live video frames.
- **OpenCV Interface:** Captures webcam video, processes image frames, and displays detection results.
- **Flask Server:** Handles backend processing, video streaming, and communication between the frontend and detection engine.
- **Frontend Interface:** Developed using HTML, CSS, Bootstrap, and JavaScript to provide an interactive web dashboard.
- **PyTorch Framework:**Executes the trained YOLOv8 deep learning model with GPU acceleration when available.
- **Voice Assistance Module:**Uses the pyttsx3 library to generate real-time voice alerts corresponding to detected traffic signs.

Chapter 6

SYSTEM DESIGN

6.1 Modules

The RoadSense AI system is divided into several functional modules to ensure efficient real-time traffic sign detection, driver assistance, and performance monitoring.

1. **User Interface Module**

    - **Accepts:** User requests through the web browser.
    - **Action:** Provides an interactive dashboard using the Flask framework.
    - **Output:** Displays live webcam feed, detected traffic signs, confidence score, FPS, driver guidance, and detection history.
    - **Processes:** User interaction, live video streaming, and dashboard visualization.

2. **Camera Capture Module**

    - **Accepts:** Live video from the webcam.
    - **Action:** Continuously captures image frames.
    - **Output:** Real-time video frames forwarded to the image processing module.
    - **Processes:** Webcam initialization, frame acquisition, and continuous video capture.

6.2 System Architecture

The RoadSense AI system follows a layered architecture to provide efficient, accurate, and real-time traffic sign detection and driver assistance.

- **Presentation Layer:** Provides the web-based dashboard displaying live video, detected traffic signs, driver assistance messages, confidence score, FPS, inference time, and GPU information.
- **Application Layer:** Handles traffic sign detection, driver assistance logic, voice alert generation, and communication between the frontend and backend.
- **Deep Learning Layer:** Utilizes the trained YOLOv8 model to detect and classify traffic signs from live webcam video in real time.
- **Data Layer:** Uses OpenCV for image acquisition, frame processing, and bounding box visualization before displaying the output.
- **Hardware Layer** Consists of the webcam for video capture and an NVIDIA GPU (optional) for accelerating deep learning inference, enabling high-speed real-time performance.

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6.3 Use Case Diagram

Use case diagrams represent the interactions between system actors and system functions.

- **User (Driver)**
  - Start Application
  - Open Camera
  - Capture Live Frame
  - Detect Traffic Sign
  - Recognize Traffic Sign
  - Calculate Confidence Score
  - Display Detection Result
  - Generate Driver Message
  - Generate Voice Alert
  - Show Driver Assistance
  - Display FPS and Inference Time
  - View Detection History
  - Stop Application
- **AI Detection System**
  - Start Application
  - Open Camera
  - Capture Live Frame
  - Detect Traffic Sign
  - Recognize Traffic Sign

- Calculate Confidence Score
- Display Detection Result
- Generate Driver Message
- Generate Voice Alert
- Show Driver Assistance
- Display FPS and Inference Time
- View Detection History
- Stop Application

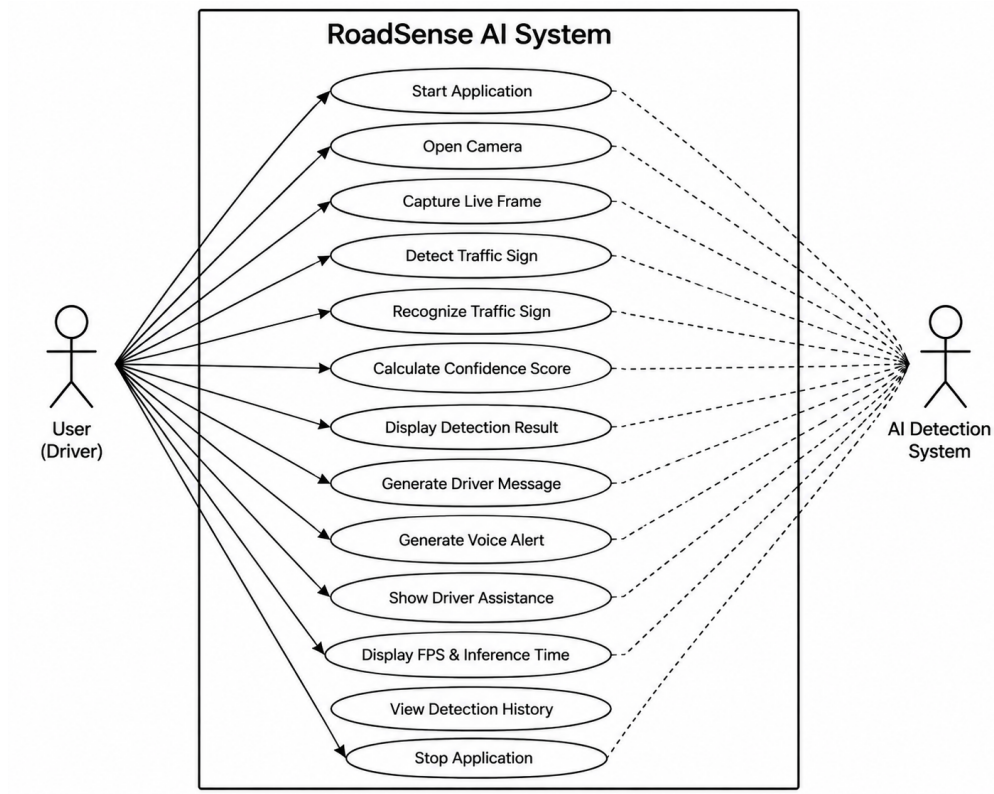


Fig. 6.1: Use Case Diagram

6.4 Sequence Diagram

6.4.1 Sequence Diagram for User(Driver)

The sequence diagram shows the interaction between the user, RoadSense AI dashboard, webcam, YOLOv8 detection model, voice assistant, and driver assistance module. The user starts the application, and the webcam continuously captures live video frames. Each frame is processed by the YOLOv8 model to detect and classify traffic signs. The detected traffic sign information, confidence score, and bounding boxes are returned to the dashboard for visualization. Based on the detected sign, the driver assistance module generates the appropriate safety message, while the voice assistant announces the detected traffic sign. The dashboard continuously updates the live camera feed, detection statistics, FPS, inference time, and GPU information until the user closes the application.

1. User opens RoadSense AI.
2. Flask server starts.
3. Webcam captures live video.
4. Frame is sent to YOLOv8.
5. YOLO detects traffic sign.
6. Detection result returned.
7. Dashboard displays sign.
8. Voice Assistant announces sign.
9. User continues driving.

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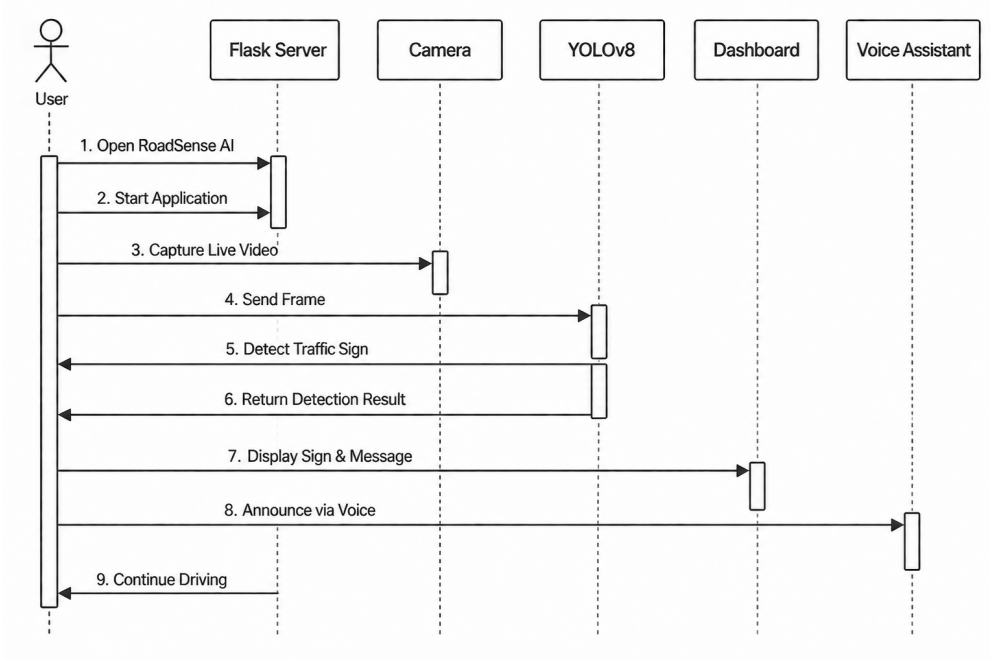


Fig. 6.2: Sequence Diagram for User(Driver)

6.4.2 Sequence Diagram for AI Detection System

The sequence diagram illustrates how the AI detection system manages the real-time traffic sign detection process. It shows the interaction between the webcam, YOLOv8 detection model, driver assistance module, voice assistant, and the RoadSense AI dashboard to detect, recognize, and display traffic signs while providing real-time visual and voice alerts to the user.

1. Capture Video Frames
2. Process Image using YOLOv8
3. Detect Traffic Signs
4. Classify Traffic Signs
5. Draw Bounding Boxes
6. Generate Voice Alert
7. Update Dashboard

8. Display Detection Statistics

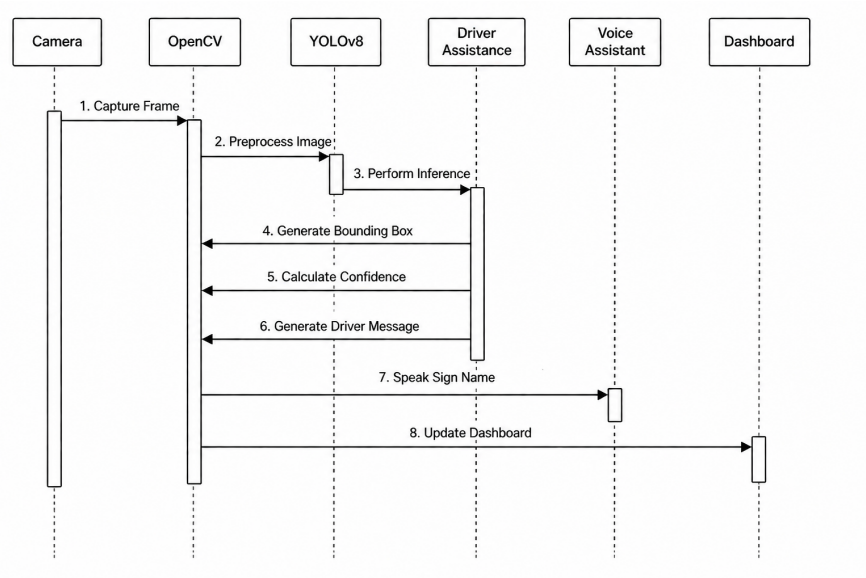


Fig. 6.3: Sequence Diagram for AI Detection System

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6.5 ER Diagram

The ER diagram represents the data structure of the RoadSense AI system.

- Defines the relationships between entities such as User, Webcam, YOLOv8 Model, Traffic Sign, Detection Result, and Voice Assistant.
- Each webcam frame is processed by the YOLOv8 model to identify and classify traffic signs in real time.
- Detection results are linked with the corresponding traffic sign, confidence score, driver assistance message, and voice alert to provide accurate real-time information to the user.

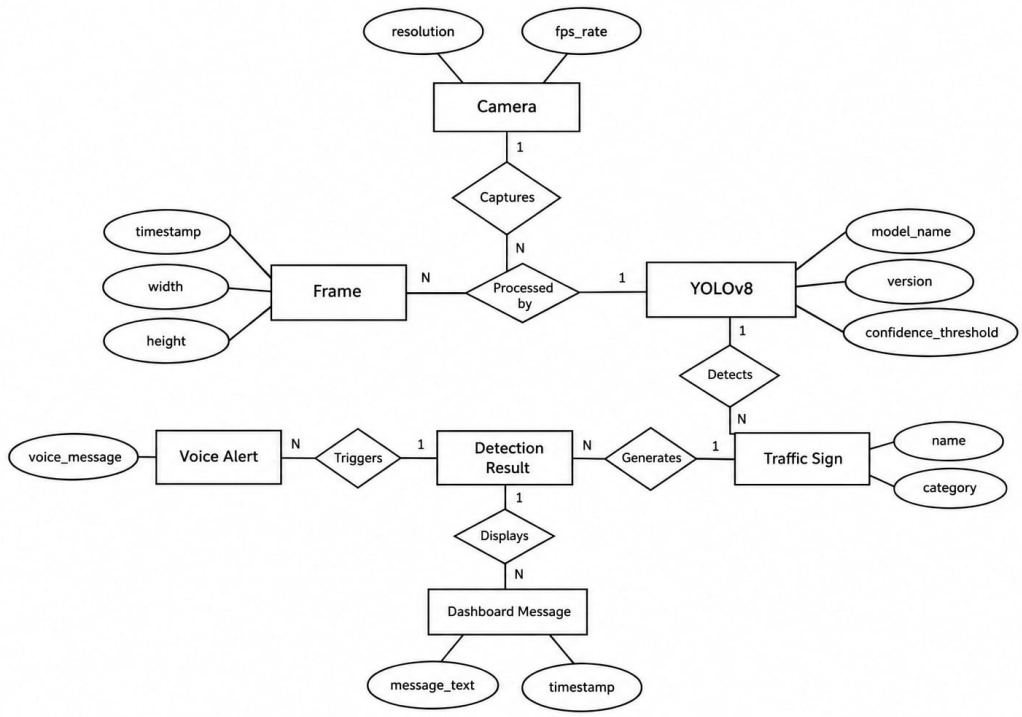


Fig. 6.4: ER Diagram for RoadSense AI system

6.6 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the RoadSense AI system.

- Shows the flow of information between the webcam, YOLOv8 detection model, voice assistant, driver assistance module, and dashboard.
- Identifies data transformations such as image capture, traffic sign detection, sign classification, and voice alert generation.
- Demonstrates how live video frames are processed, detected, classified, and displayed with real-time driver assistance information.

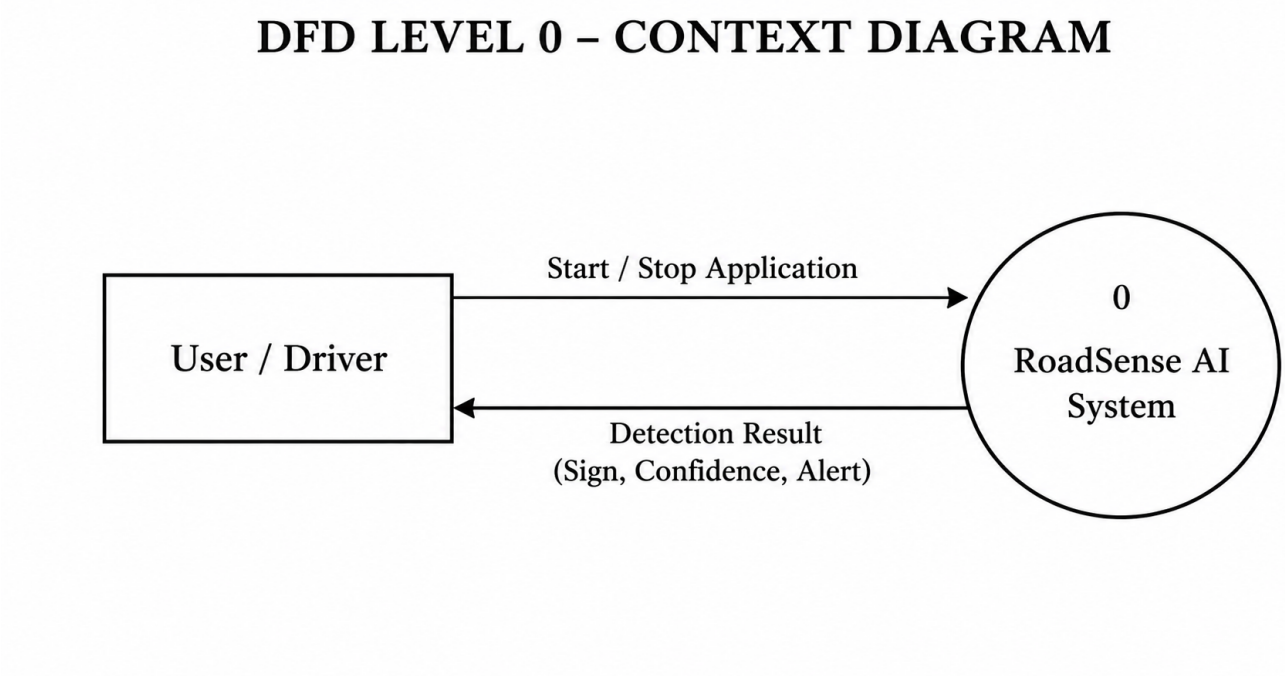


Fig. 6.5: DFD Level 0

6.6.1 DFD Level 0

6.6.2 DFD Level 1 (User(Driver))

6.6.3 DFD Level 1 (AI Detection System)

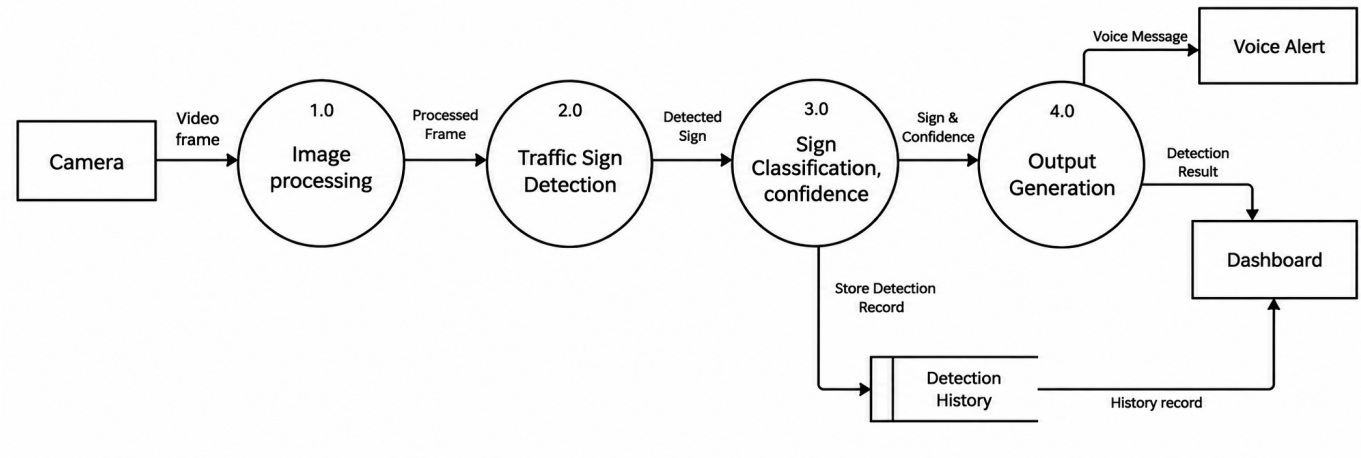


Fig. 6.6: DFD Level 1 - User(Driver)

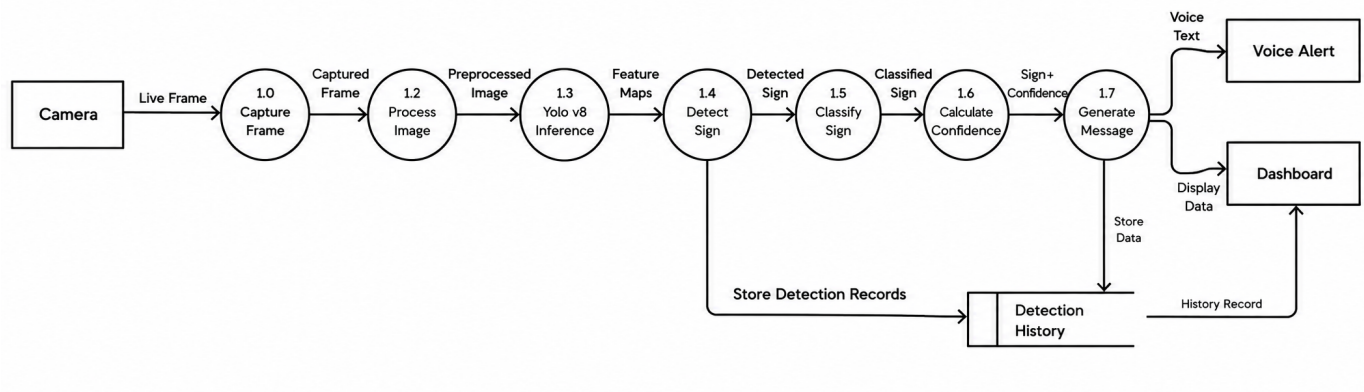


Fig. 6.7: DFD Level 2 - AI Detection System

Chapter 7

IMPLEMENTATION

7.1 Introduction

Implementation is an important phase in the software development life cycle where the designed system is transformed into a working application. In the RoadSense AI project, this phase focuses on developing a real-time traffic sign detection and recognition system using the YOLOv8 deep learning algorithm. The system captures live video from a webcam, detects and classifies traffic signs, displays detection results with confidence scores, and provides real-time driver assistance through visual notifications and voice alerts. The implementation integrates computer vision, deep learning, and web technologies to ensure accurate detection, fast processing, and an interactive user experience.

7.2 Frontend Implementation

The frontend of the system is developed using HTML, CSS, JavaScript, and Bootstrap, providing an interactive and responsive dashboard for real-time traffic sign monitoring.

- **Live Camera Interface:** Displays the webcam video stream with detected traffic signs, bounding boxes, confidence scores, and real-time detection results.
- **Dashboard Interface:** Shows traffic sign information, driver assistance messages, FPS, inference time, GPU information, and system status in an organized and user-friendly layout.

7.3 Backend Implementation

The backend of the system is implemented using **Python and Flask.** It manages server-side operations, processes webcam frames, performs traffic sign detection using the YOLOv8 model, and communicates with the frontend.

- **Camera Management:**Captures live video frames from the webcam using OpenCV.
- **Traffic Sign Detection:** Processes each frame using the YOLOv8 model to detect and classify traffic signs with high accuracy.
- **Driver Assistance:** Generates appropriate safety instructions based on the detected traffic sign.
- **Voice Alert System:** Uses the pyttsx3 text-to-speech engine to announce detected traffic signs in real time.
- **Dashboard Update:** Sends detection information such as traffic sign name, confidence score, FPS, inference time, and GPU details to the dashboard through Flask APIs.

7.4 Modules Implementation (Summary)

- **Live Camera Module:** Captures continuous video frames from the webcam for real-time processing.
- **Traffic Sign Detection Module:** Uses the trained YOLOv8 model to detect and classify traffic signs from the live video stream.
- **Driver Assistance Module:**Generates safety instructions based on the detected traffic sign to assist the driver.
- **Voice Alert Module:** Administrators verify election data and publish the final results after vote counting.

Chapter 8

RESULT AND ANALYSIS

8.1 Screenshots

8.1.1 InterFace

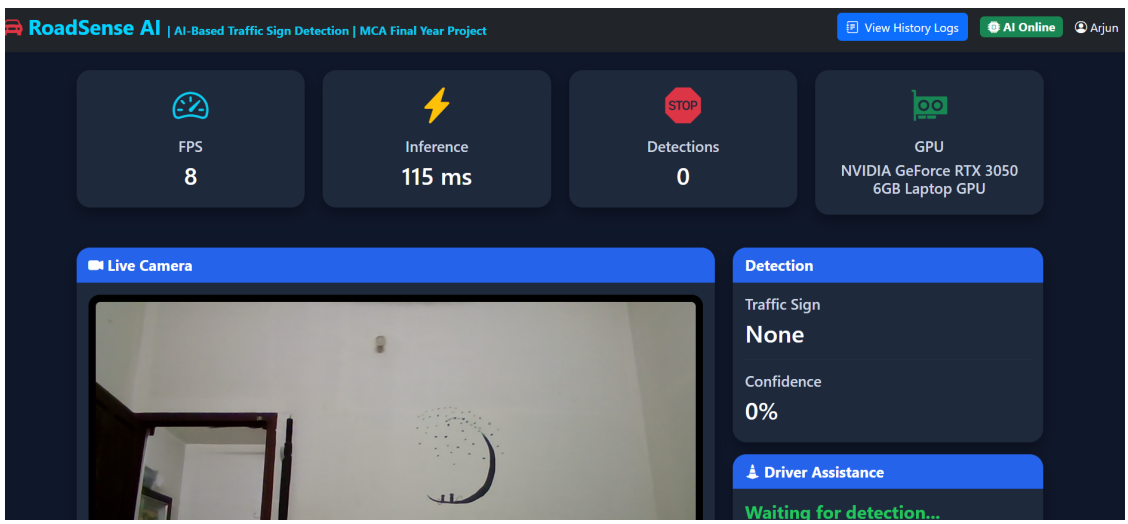


Fig. 8.1: User Interface

8.1.2 Live Detect

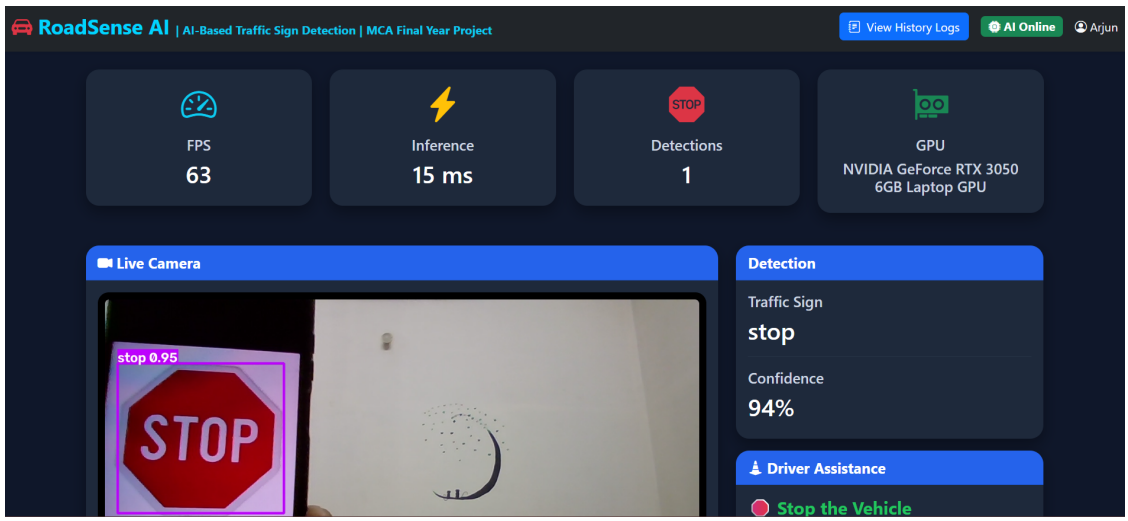


Fig. 8.2: Live Detection of Sign

8.1.3 Round Bound Ai

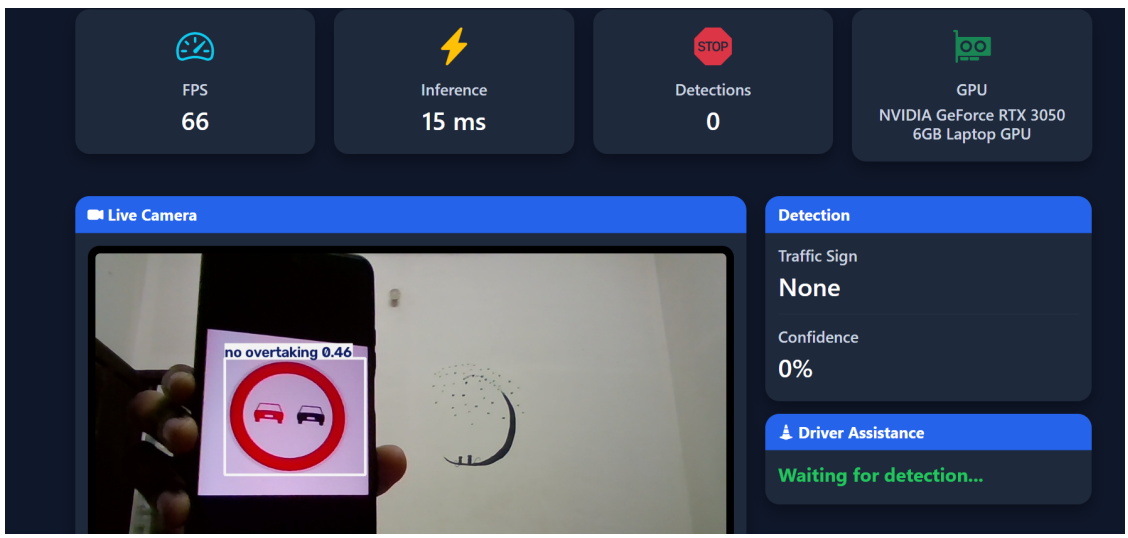


Fig. 8.3: AI Agent

8.1.4 Result Detection

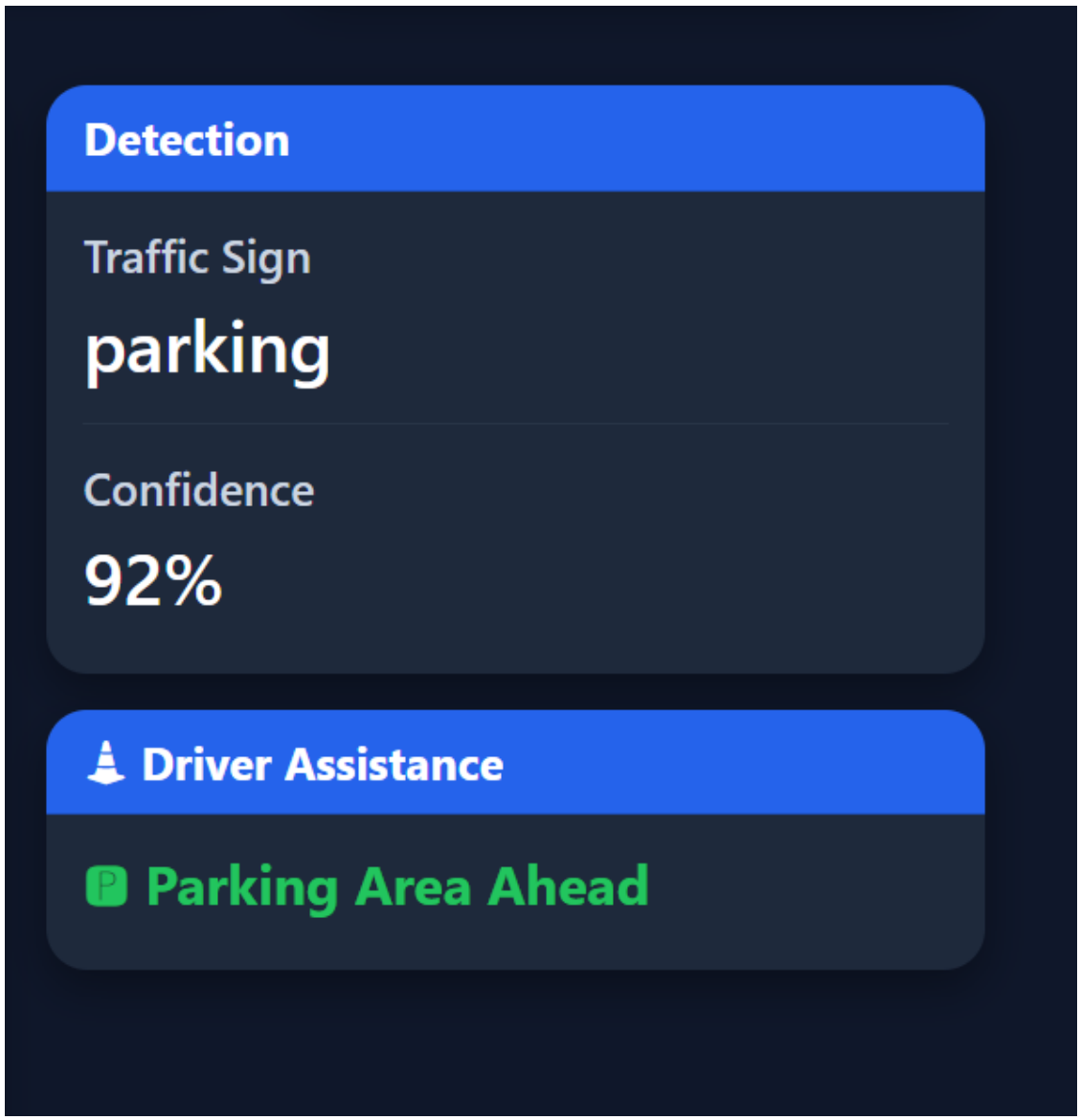


Fig. 8.4: Result Detection

8.1.5 Inference

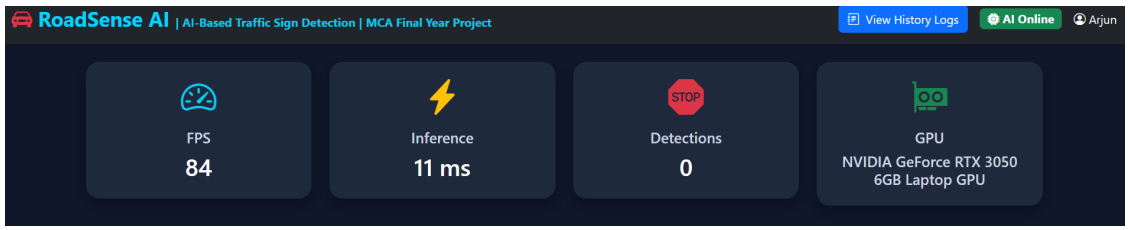


Fig. 8.5: Inference

8.1.6 Detection History

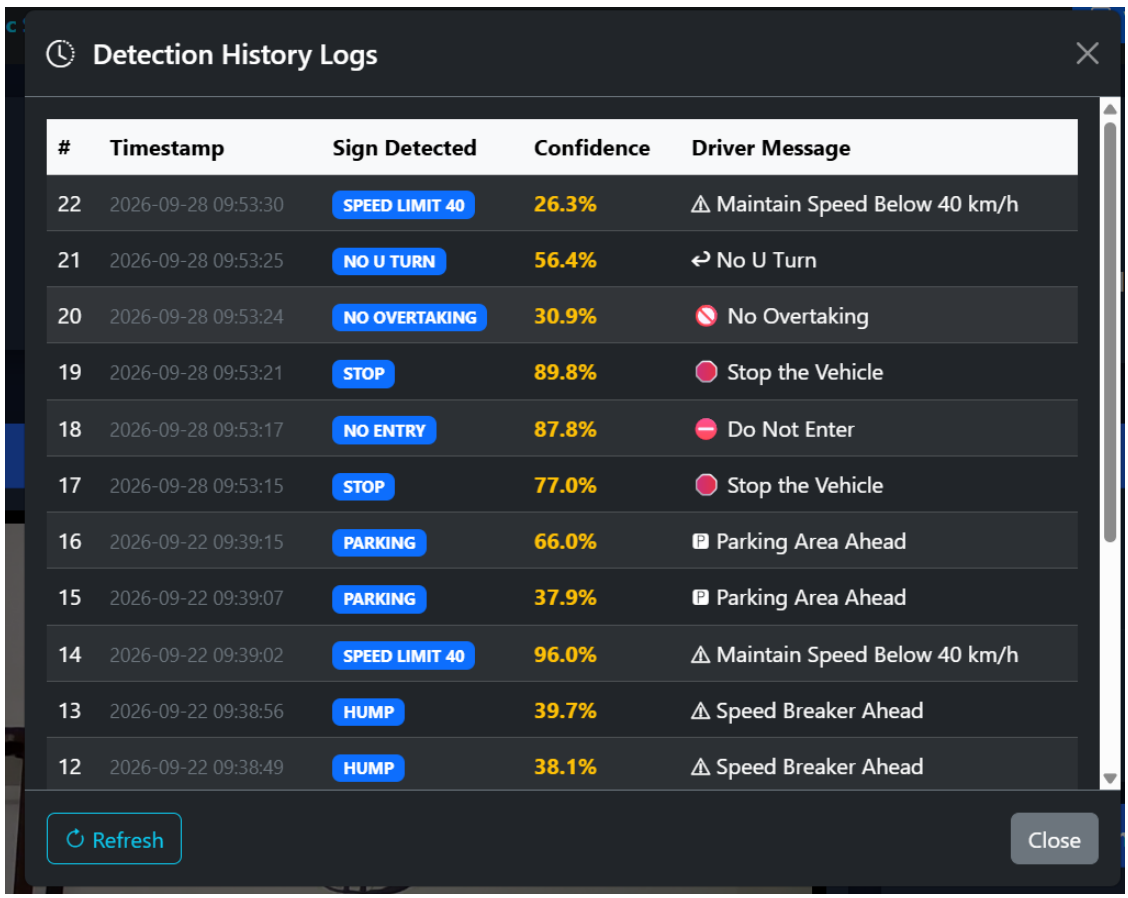


Fig. 8.6: History

Chapter 9

SAMPLE CODES

9.1 Index

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>RoadSense AI - Detection History</title>
    <!-- Bootstrap 5 CSS -->
    <link href="https://cdn.jsdelivr.net/style.css" rel="stylesheet">
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="bg-dark text-white py-4">

    <div class="container">
        <!-- Header -->
        <div class="d-flex justify-content-between align-items-center mb-4">
            <h2>🚗 RoadSense AI - Detection Logs</h2>
            <a href="/" class="btn btn-outline-light">← Back to Dashboard</a>
        </div>

        <!-- History Table -->
        <div class="card bg-secondary text-white shadow-lg">
            <div class="card-body">
                {% if records %}
                <div class="table-responsive">
                    <table class="table table-dark table-hover align-middle mb-0">
                        <thead class="table-light">
                            <tr>
                                <th>#</th>
                                <th>Timestamp</th>
```

9.2 App code

```python
def log_detection_to_db(sign_name, confidence, driver_message):
    """Safely logs a detection event to MySQL without clogging the webcam stream."""
    try:
        conn = mysql.connector.connect(**DB_CONFIG)
        cursor = conn.cursor()

        sql = """
        INSERT INTO detection_history (sign_name, confidence, driver_message)
        VALUES (%s, %s, %s)
        """

        # CRITICAL FIX: Convert PyTorch/NumPy values to native Python types (str, float
        clean_sign = str(sign_name)
        clean_confidence = round(float(confidence), 2)
        clean_message = str(driver_message)

        values = (clean_sign, clean_confidence, clean_message)

        cursor.execute(sql, values)
        conn.commit()

        cursor.close()
        conn.close()

        print(
            f"✅ [DATABASE SAVED] Sign: {clean_sign} | Conf: {clean_confidence}%"
        )
```

9.3 Javascript

```javascript
function updateDetection() {
    fetch("/detection")
        .then(response => response.json())
        .then(data => {
            // Detection Panel
            document.getElementById("sign").innerHTML = data.sign;
            document.getElementById("confidence").innerHTML = data.confidence;
            document.getElementById("action").innerHTML = data.action;

            // Statistics Cards
            document.getElementById("fps").innerHTML = data.fps;
            document.getElementById("inference").innerHTML = data.inference + " ms";
            document.getElementById("detections").innerHTML = data.detections;
            document.getElementById("gpu").innerHTML = data.gpu;
        })
        .catch(error => console.error("Error updating live detection:", error));
}

// Update live stats every 300 ms
setInterval(updateDetection, 300);

// Run immediately when the page loads
updateDetection();
```

9.4 Dataset

```yaml
train: ../train/images
val: ../valid/images
test: ../test/images

nc: 10
names: ['hump', 'no entry', 'no overtaking', 'no stopping', 'no u turn', 'parking', 'roadwork', 'roundabout', 'speed limit 40', 'stop']

roboflow:
  workspace: university-km5u7
  project: traffic-sign-detection-yolov8-awuus
  version: 8
  license: CC BY 4.0
  url: https://universe.roboflow.com/university-km5u7/traffic-sign-detection-yolov8-awuus/dataset/8
```

9.5 Train

```python
from ultralytics import YOLO
from pathlib import Path

ROOT = Path(__file__).resolve().parent.parent

def main():
    model = YOLO("yolov8n.pt")

    model.train(
        data=str(ROOT / "dataset" / "data.yaml"),
        epochs=50,
        imgsz=640,
        batch=16,
        device=0,
        workers=0,          # <-- Important change
        project="runs",
        name="RoadSenseAI"
    )

if __name__ == "__main__":
    main()
```

Chapter 10

CONCLUSION & FUTURE SCOPE

10.1 Conclusion

The development of the RoadSense AI system demonstrates the feasibility of using deep learning and YOLO-based object detection to create a real-time traffic sign detection and recognition system. The proposed system addresses important challenges in road safety by automatically identifying traffic signs from live camera input and providing immediate information to the driver. By using YOLO, the system is capable of detecting traffic signs efficiently while maintaining a suitable balance between detection accuracy and real-time performance.

The system also provides driver assistance through confidence-based detection, voice alerts, and informative messages for recognized traffic signs. The integration of YOLO with OpenCV and Flask enables continuous monitoring through a live webcam feed, while FPS and inference time provide information about the system's real-time performance. Overall, the project demonstrates that combining deep learning-based traffic sign detection with real-time driver alerts can improve traffic sign awareness and contribute to safer driving conditions.

10.2 Future Scope

Although the proposed RoadSense AI system provides real-time traffic sign detection and driver assistance, several improvements can be implemented in the future to enhance its accuracy, reliability, and functionality. The system can be extended by training the YOLO model with larger and more diverse traffic sign datasets covering different weather, lighting, and road conditions. Additional traffic sign classes can also be included to improve the overall coverage of the system. Future developments can focus on integrating GPS and navigation systems to provide location-based traffic sign information and advanced driver assistance. The system can also be extended to detect other road objects such as vehicles, pedestrians, lane markings, and traffic signals. More advanced YOLO models and optimization techniques can be used to improve detection accuracy while reducing inference time. In addition, the system can be deployed on edge devices or embedded systems for practical use in vehicles. With these advancements, RoadSense AI can evolve into a more accurate, efficient, and comprehensive intelligent road safety and driver assistance system.

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